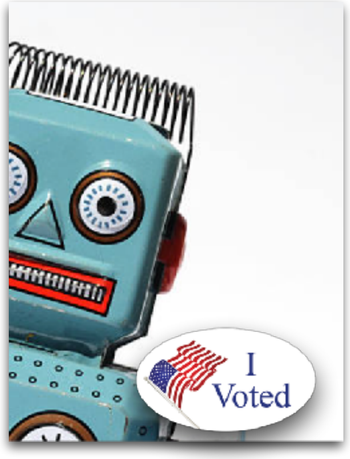
Modeling Democracy

Lecture 17 - **So can we get to fairness after all??**

—**the redistricting case.**



Moon Duchin, Winter 2026 — Wed Mar 4

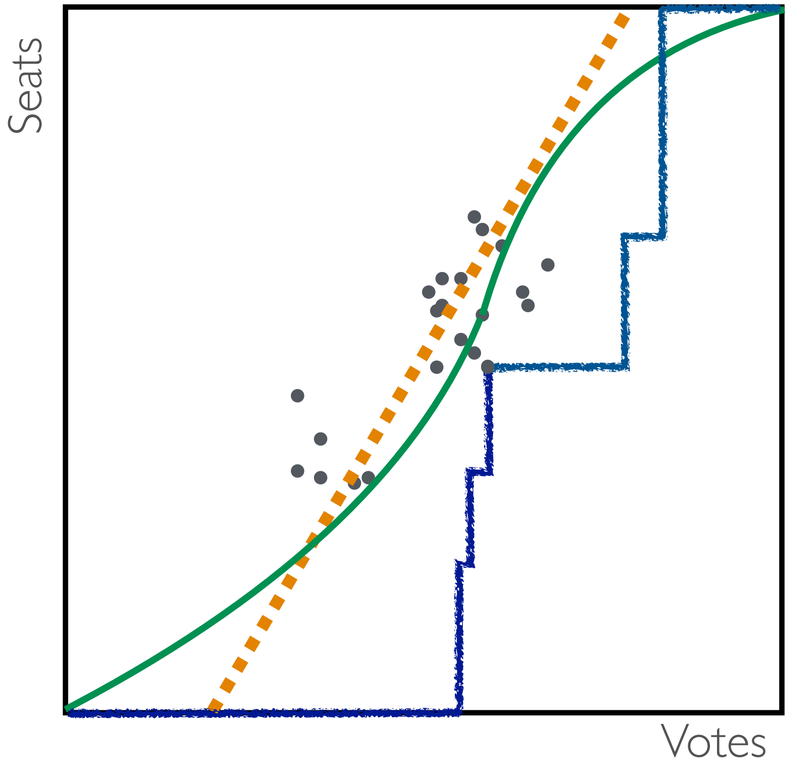
Recall: seats vs votes

Pairing a plan and a vote pattern gives a **point**

So if you have a source of districting plans (like an ensemble algorithm) and a source of election data, you will get a **scatter**

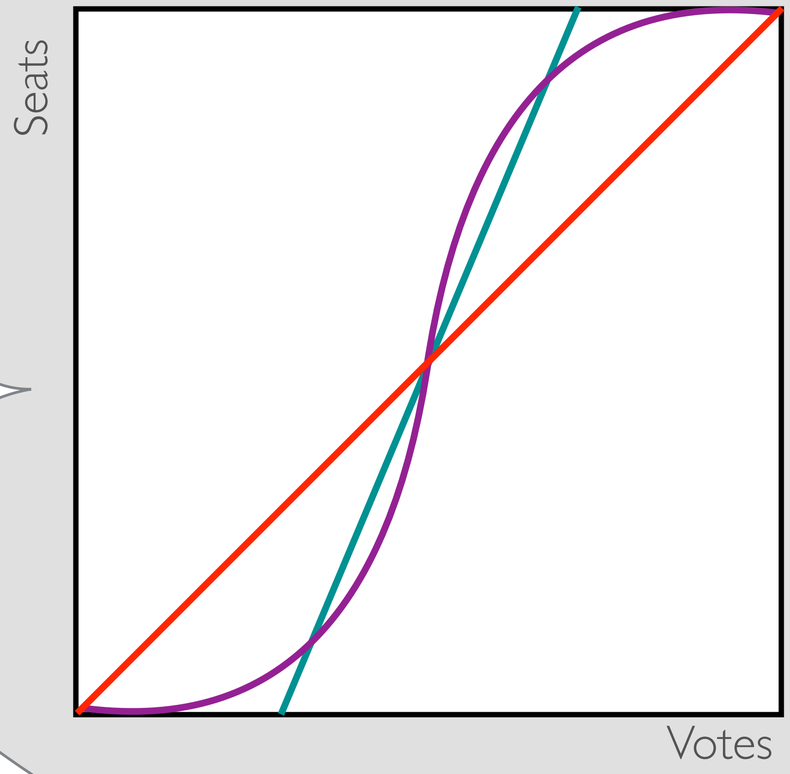
- Can fit a line
- Can fit a curve
- Can build out a step function

Some scoring ideas are based on comparing each of these to the "ideals"



- Proportionality: ideal S=V
- "Efficiency gap": ideal 2S-4V=1
- Partisan symmetry: ideal S(V) symmetric

Plot for 2-party case



Hidden detail: What election data?

- This is a sneaky huge issue! And it can matter.

- Exogenous/endogenous: use **same** type/level of election as the contests you are analyzing? Most data scientists prefer exogenous elections that cover the whole jurisdiction (e.g., statewide contests like President, U.S. Senate to analyze congressional/legislative maps)

- "Election index": blend of elections — but how?
    - just add or average *counts* of votes by precinct over a set of contests
    - average *shares* of votes by precinct over a set of contests
    - set up a regression with variables for factors like incumbency advantage, uncontested races, etc and "learn" a best combination of set of elections

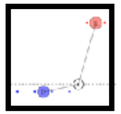
- my claim: none of these works very well to produce an index that has properties resembling the class of elections you are combining

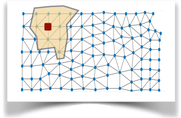
every metric draws on a different idea of the "state of nature"

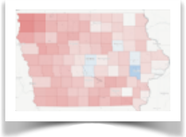
Group I: votes vs. outcomes

Group II: vote shares alone

Group III: votes and geography







metric	notable proponents	target	ideal situation	measurement
Proportionality	traditional	$S = V$	seat share equals vote share	$disprop = S - V$
Efficiency gap	Steph.–McGhee, Cain, Johnston et al [12, 10]	$\star$ and $\Delta S = 2\Delta V$	double-bonus (seat share rises by two points for every point of vote share)	$SEG = 2V - S - 1/2$
Mean-median (partisan symmetry)	Grofman, Katz–King et al., Wang, etc [8, 11, 14]	partisan swing hits $\star$	median district reflects statewide vote	MM is vote displacement from $\star$
Partisan bias (partisan symmetry)	Gelman–King [7]	partisan swing hits $\star$	half of the votes "deserves" half of the seats	PB is seats displacement from $\star$
Declination	Warrington [15]	$\frac{ADM}{\#D} = \frac{ARM}{\#R}$	each party should have same average win margin over number of wins	δ measures an angle between R win centroid, D win centroid, and a central point
Ensemble mean/median/mode	"in the air"	average / 50th percentile / most frequent S value	plans should measure similar to typical neutrally generated plan	outlier level
Partisan dislocation	DeFord-Eubank-Rodden [5]	$S =$ what would be obtained from personal districts	partisanship of voters' district-sized neighborhood matches with their district	$AAPD$ is difference
Artificial partisan advantage	Eguia [6]	$S =$ what would be obtained from pop-weighted counties	seat share equals population share in counties won by party	APA is difference

Principles for fairness metrics

- Be clear about **what** you're trying to measure
- Metric should be **sensitive to what matters**, not sensitive to what doesn't
- For a reasonable regulation, rule should be **achievable** by good-government actors and should include room for exceptional cases
- Induced **incentives** should align with policy goals
- Should admit a healthy **procedure**
- Should be robustly tested against **ground truth** (some instances with agreed labels) and should stand up to sensitivity analysis

The anthropologist and STS scholar Marilyn Strathern famously riffed that "**When a measure becomes a target, it ceases to be a good measure**."

To take this adage seriously, a real-world analysis must consider metrics in reductionist and adversarial conditions.

*riffing on monetary policy quote by Charles Goodhart: ''Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes.''

Partisan symmetry fails these tests

DeFord, Dhamankar, Duchin, Gupta, McPike, Schoenbach, Sim

Theorem 3 (Partisan Symmetry Characterization). *Given k districts with vote shares $\mathbf{v}$, jump vector $\mathbf{j}$, and gap vector δ, the following are equivalent:*

$$\mathrm{PG}(\mathbf{v}) = 0 \qquad \text{(Partisan Symmetry Standard)}$$

$$j_i + j_{k+1-i} - 1 = 0 \quad \forall i \qquad \text{(jumps)}$$

$$\frac{1}{2}(v_i + v_{k+1-i}) = \overline{v} \quad \forall i \qquad \text{(mean vote)}$$

$$\frac{1}{2}(v_i + v_{k+1-i}) = v_{\text{med}} \quad \forall i \qquad \text{(median vote)}$$

$$\delta_i = \delta_{k-i} \quad \forall i \qquad \text{(gaps)}$$

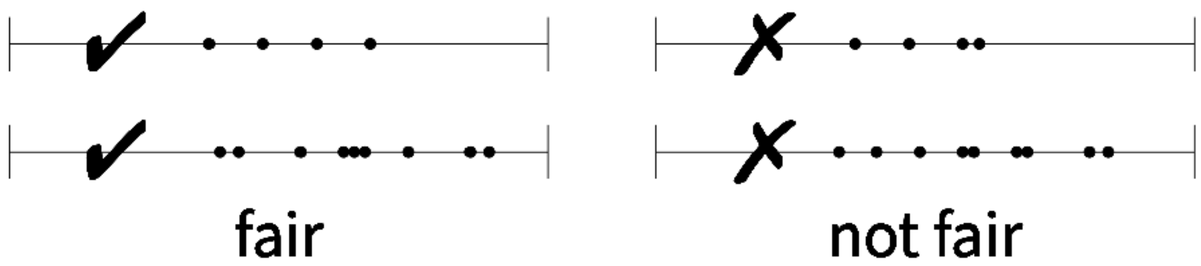


Figure 3. Four election outcomes, shown as vote shares by district. On the left-hand side, the v_i are symmetric about their center, so all partisan symmetry scores are perfect. On the right-hand side, non-symmetric outcomes. The partisan symmetry standard can be eyeballed by a glance at the vote shares in the districts.

Partisan symmetry scores are perfect IF AND ONLY IF the vote shares by district are symmetrically spaced on the number line

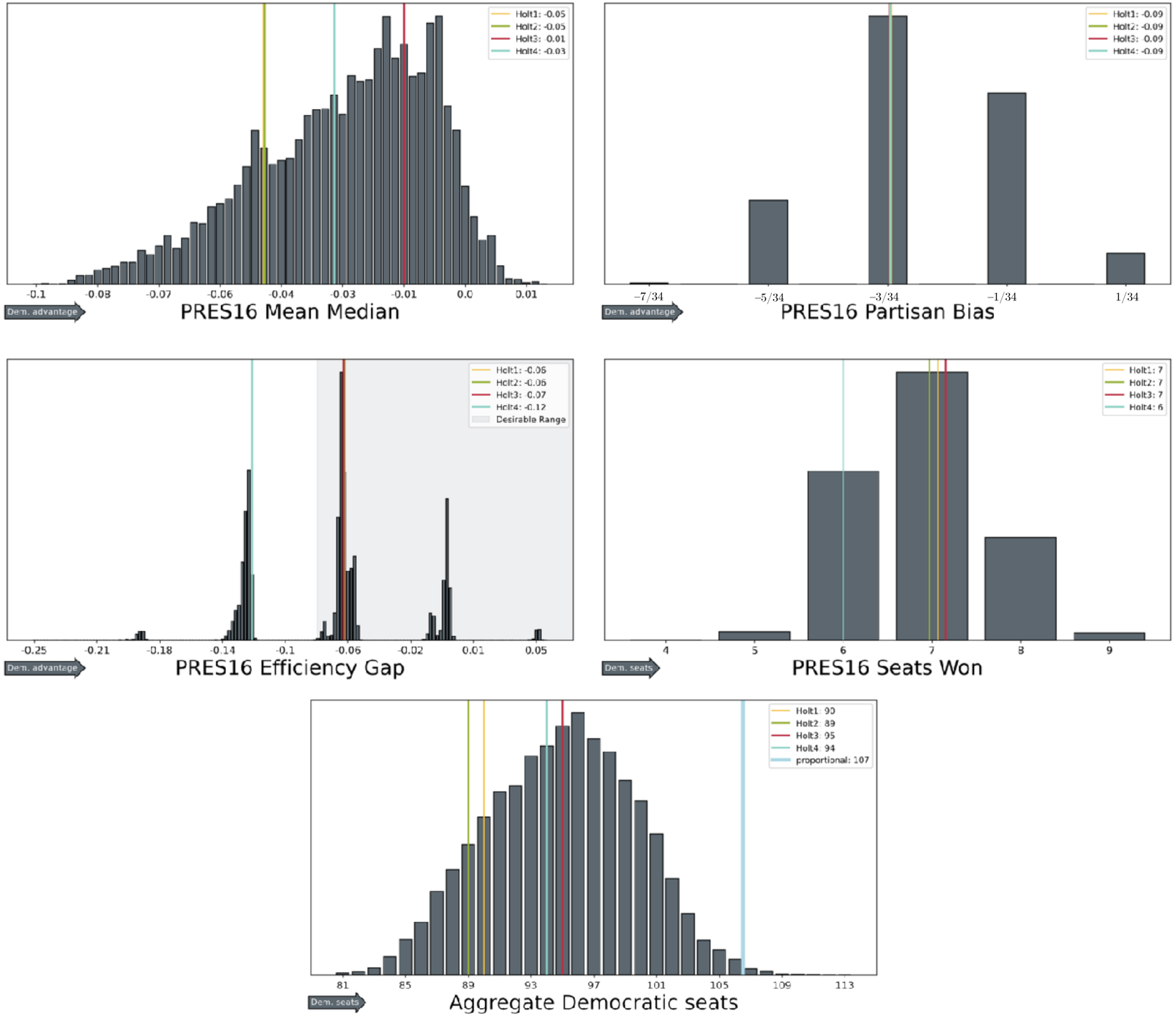
- clear on what is measured ❌
- sensitive to what matters ❌
- good scores achievable ✅
- creates good incentives ❌
- allows healthy procedure 🆗
- gets ground truth right ❌

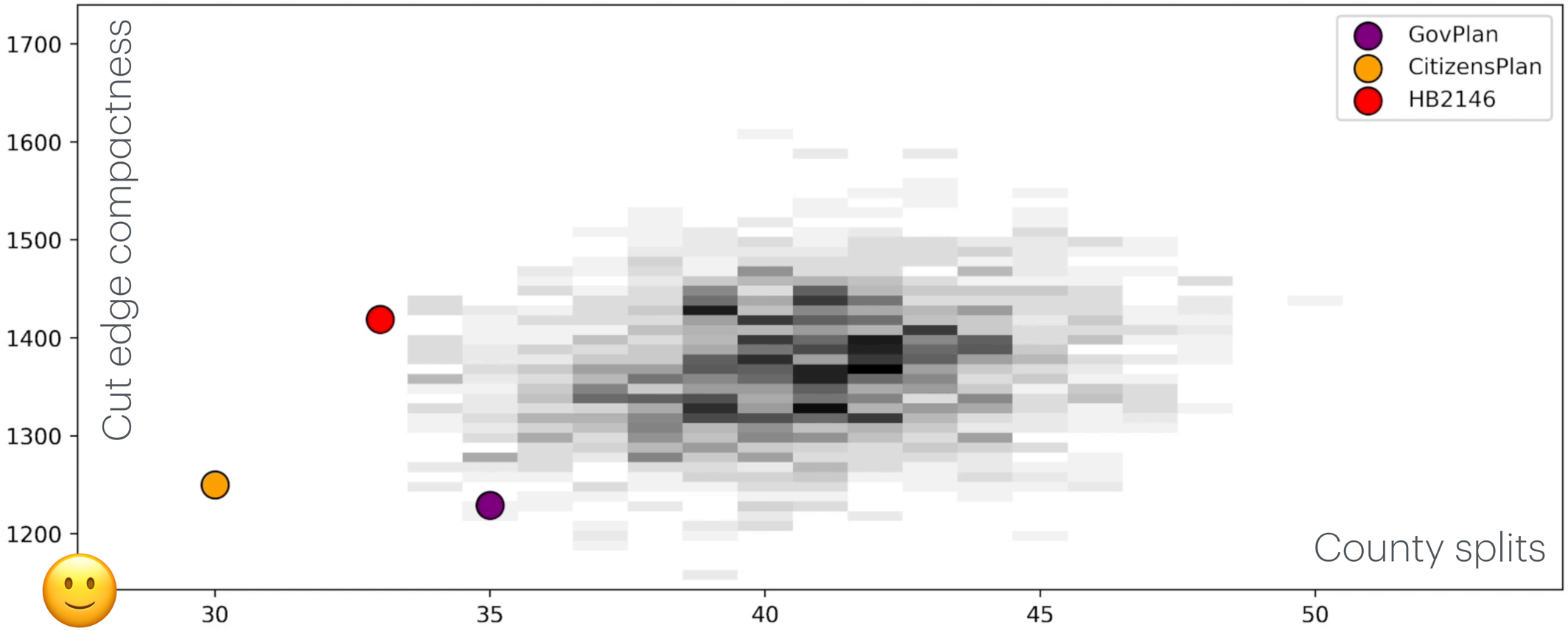
let’s look at some empirics

Neutral plans don't always give scores centered at 0

Neutral plans often fail naive bright-line tests

Should typical be elevated to ideal? Or are there scores where it's worth it to fight the tendencies of neutral plans?




GovPlan
CitizensPlan
HB2146
Cut edge compactness
County splits
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1500
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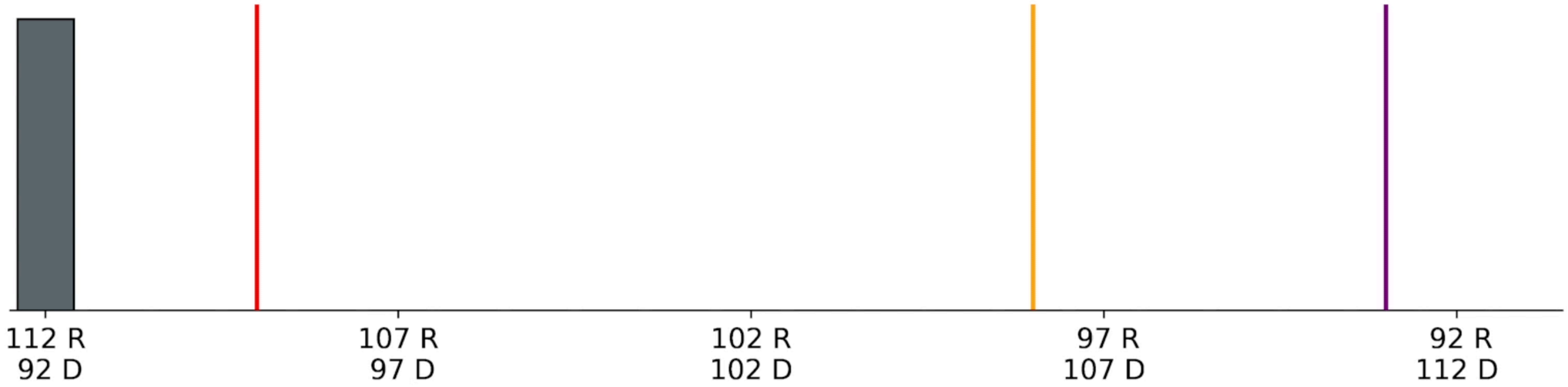

112 R
92 D
107 R
97 D
102 R
102 D
97 R
107 D
92 R
112 D

Let's consider **proportionality** metric for districts

- Use an election index or a serial test

- Define (signed) disproportionality for a single election as S-V, so that positive outputs flag advantage for the point-of-view party.

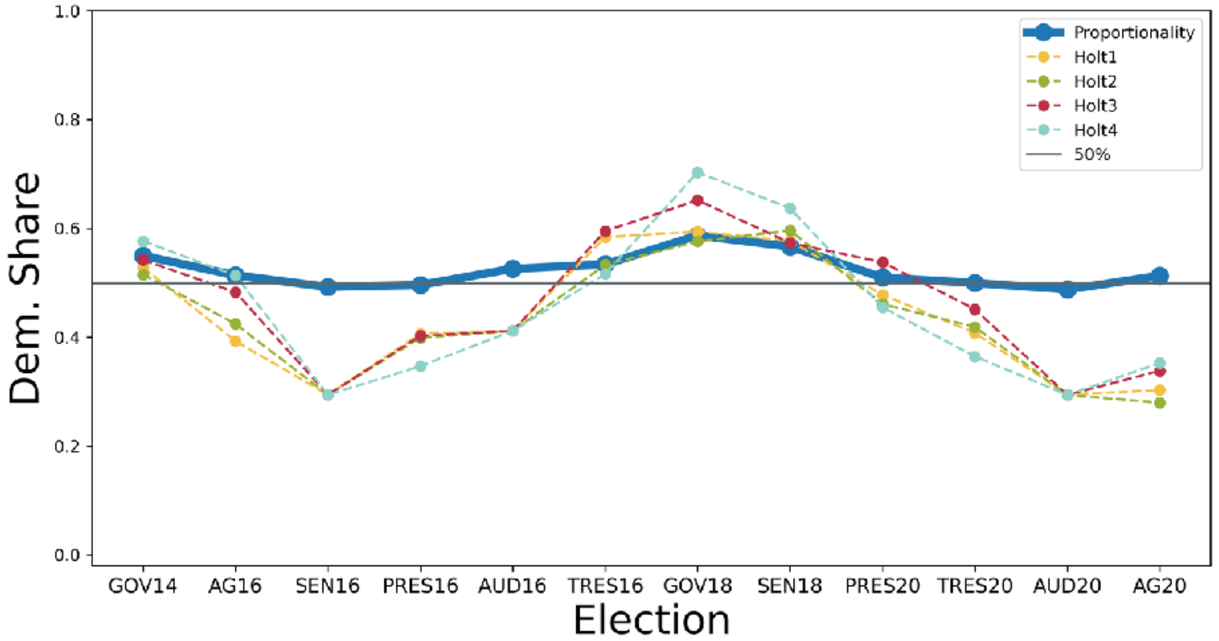
- then can define for serial dataset $\mathscr{E} = \{E_i\}$ via

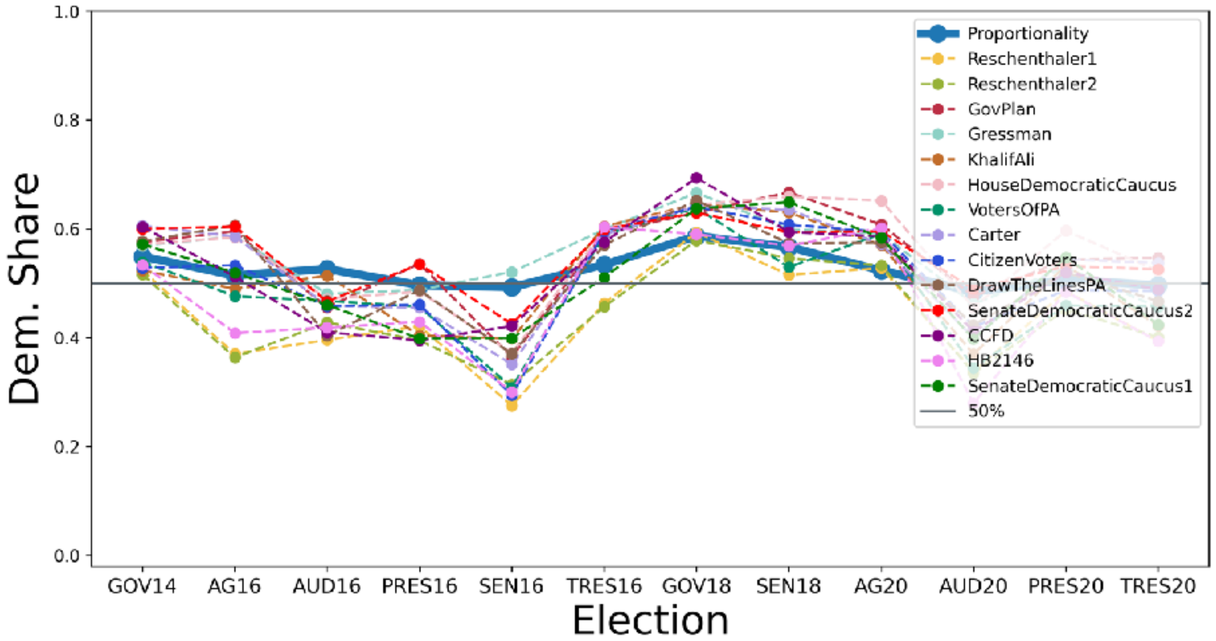
$$disprop(P, \mathscr{E}) = \sum S_i - V_i \qquad \text{or} \qquad \overline{disprop}(P, \mathscr{E}) = \sum |S_i - V_i|$$

- the first gives a good score to a plan with **responsive proportionality**, the second rewards **consistent proportionality**

Visualizing proportionality

- Fat blue line shows the **vote** share by election
- other colored lines show **seat** shares under various districting plans
- highs and lows cancel out $\leftrightarrow disprop \approx 0$
- proportional every time $\leftrightarrow \overline{disprop} \approx 0$





So what might a good rule look like?

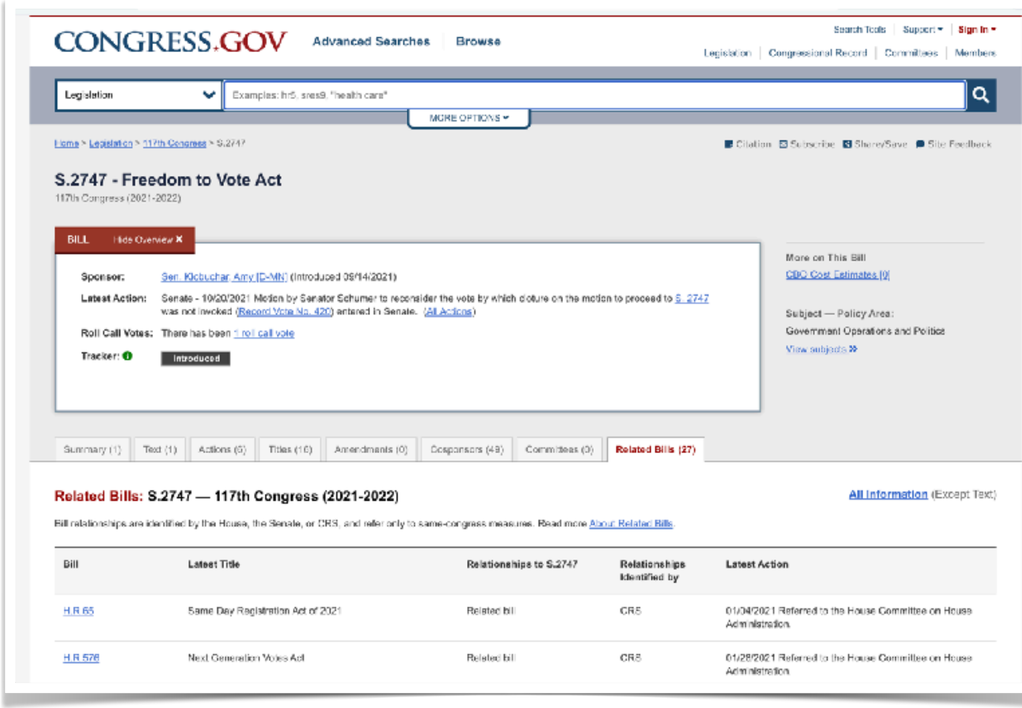
- My best attempt can be found in S.2747 — Freedom to Vote Act — 117th Cong (2021-2022). Here's a link. Also see analytical supporting paper in The Forum.

- Use four statewide elections: two last President elections, two last Senate.

- Choose your fairness target for S, such as what is given by Proportionality (S=V) or Efficiency Gap (S=2V-S-1/2).

- Be within .07 or 1 seat at least three times out of four.

- Otherwise, **rebuttable presumption** of gerrymandering. (Penalty box!)

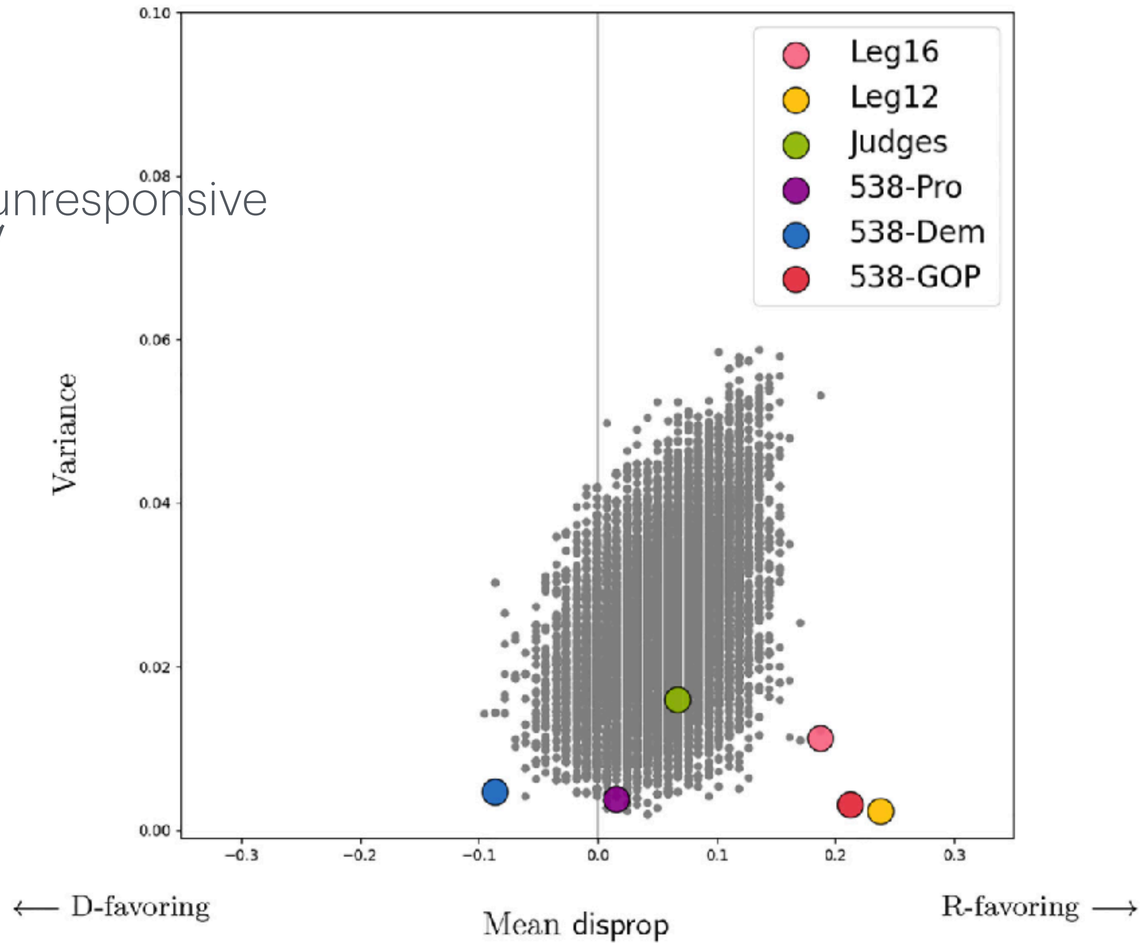


Let's see this in a real case: NC

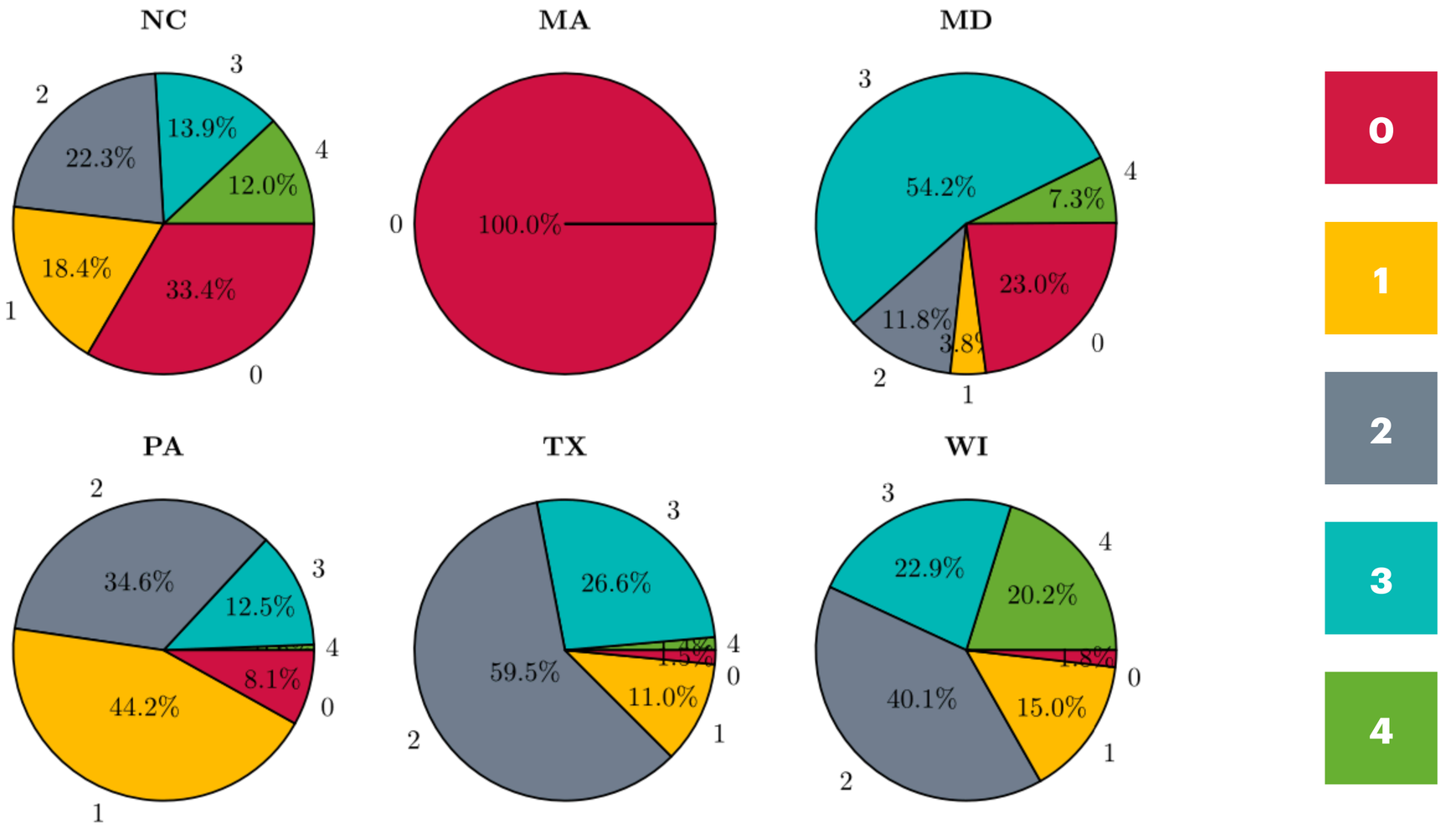
targets can be very different

unresponsive

	Election	Gov08	Sen08	Sen10	Gov12	Pres12	Sen14	Pres16	Sen16	Gov16
	R vote share	**0.483**	**0.457**	**0.560**	**0.559**	**0.511**	**0.508**	**0.520**	**0.530**	**0.500**
targets	Proportional	6.3	5.9	7.3	7.3	6.6	6.6	6.8	6.9	6.5
	EG = 0	6.0	5.4	8.1	8.0	6.8	6.7	7.0	7.3	6.5
	Ensemble mean	4.3	3.0	10.3	9.7	7.3	7.6	7.8	8.5	7.3
Leg12	R seats	8	10	10	10	10	10	10	10	10
	R seat share	0.615	0.769	0.769	0.769	0.769	0.769	0.769	0.769	0.769
	disprop	0.132	0.312	0.209	0.211	0.258	0.261	0.249	0.239	0.270
Leg16	R seats	7	5	10	10	10	10	10	10	10
	R seat share	0.538	0.385	0.769	0.769	0.769	0.769	0.769	0.769	0.769
	disprop	0.055	−0.072	0.209	0.211	0.258	0.261	0.249	0.239	0.270
Judges	R seats	4	4	9	9	8	9	8	9	8
	R seat share	0.308	0.308	0.692	0.692	0.615	0.692	0.615	0.692	0.615
	disprop	−0.175	−0.149	0.132	0.134	0.105	0.184	0.096	0.162	0.116
538Pro	R seats	5	6	9	7	7	7	7	7	7
	R seat share	0.385	0.462	0.692	0.538	0.538	0.538	0.538	0.538	0.538
	disprop	−0.098	0.005	0.132	−0.020	0.028	0.030	0.019	0.008	0.039



At least 10% of neutral plans pass the test in most states



State	Control	FTV elections				P/F
NC 2012	R	+	+	+	+	F
NC 2016	R	+	+	+	+	F
MA	D	–	–	–	–	F
MD	D	–	–	–	–	F
PA 2011	R	+	✓	+	+	F
PA 2018	R	+	+	+	+	F
TX	R	+	+	+	+	F
WI	R	+	+	+	+	F

Court-labeled gerrymanders in NC, MD, PA, TX, and WI all **fail** initial screen. So does MA.

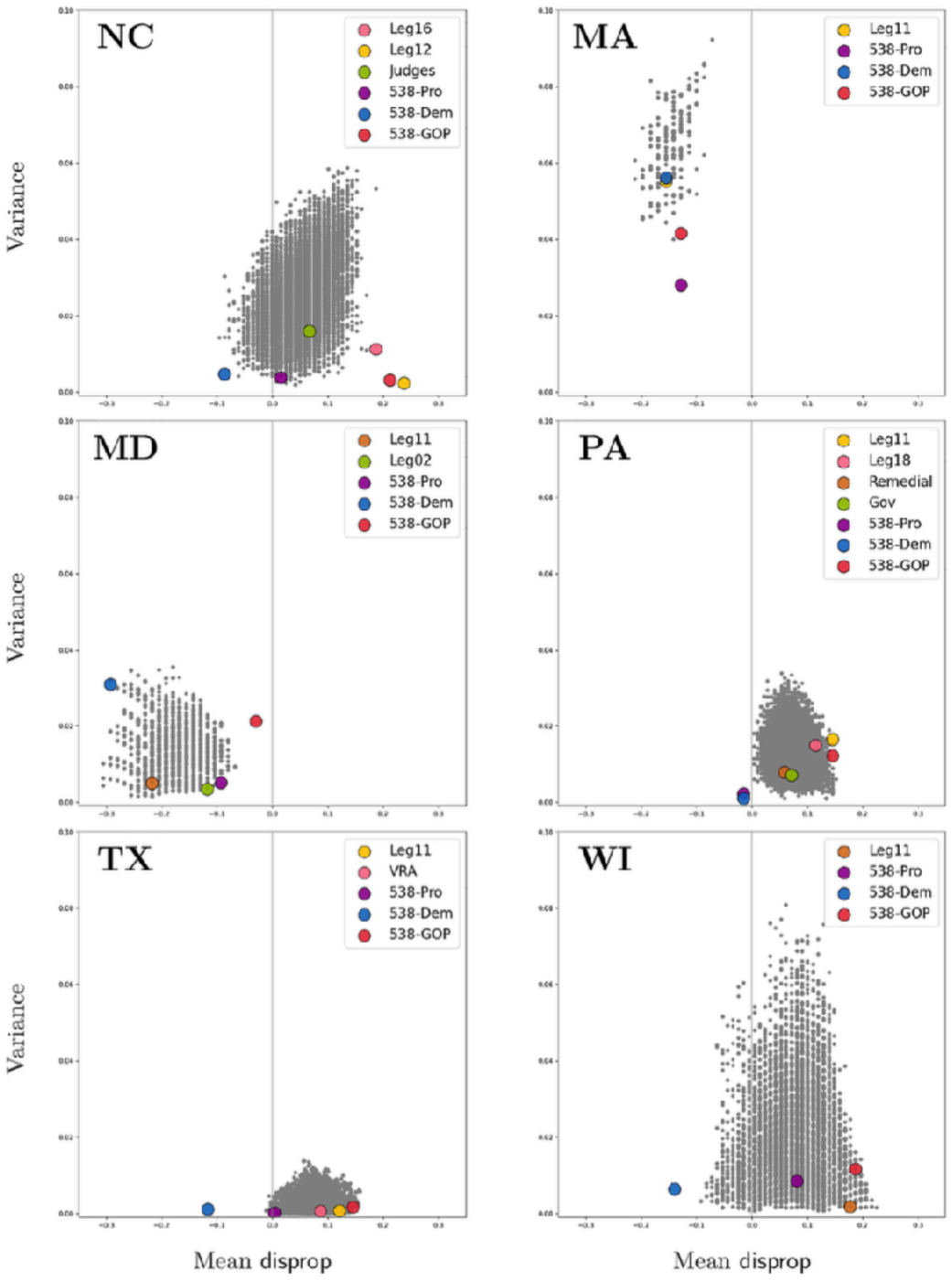
State	Alternative	FTV elections				P/F	Rebut?
NC	Judges	+	+	+	+	F	No
	538-Pro	✓	✓	✓	✓	P	
	Ensemble25	✓	✓	✓	+	P	
	Ensemble10	✓	✓	✓	✓	P	
MA	538-Pro	–	–	–	–	F	Strong
	Ensemble25	–	–	–	–	F	
	Ensemble10	–	–	–	–	F	
MD	2002	✓	✓	✓	✓	P	No
	538-Pro	✓	✓	✓	✓	P	
	Ensemble25	✓	✓	✓	–	P	
	Ensemble10	✓	✓	✓	–	P	
PA	Remedial	✓	+	✓	✓	P	Weak
	538-Pro	✓	✓	✓	✓	P	
	Gov (D)	+	✓	+	+	F	
	Ensemble25	✓	✓	+	+	F	
	Ensemble10	✓	✓	✓	+	P	
TX	VRA	✓	✓	+	+	F	No
	538-Pro	✓	✓	✓	✓	P	
	Ensemble25	✓	+	✓	✓	P	
	Ensemble10	✓	+	✓	✓	P	
WI	538-Pro	✓	✓	✓	✓	P	No
	Ensemble25	✓	✓	+	✓	P	
	Ensemble10	✓	✓	✓	✓	P	

There are good-government alternatives (and at least 10% of neutral ensemble) that **pass** screen in all states but MA.

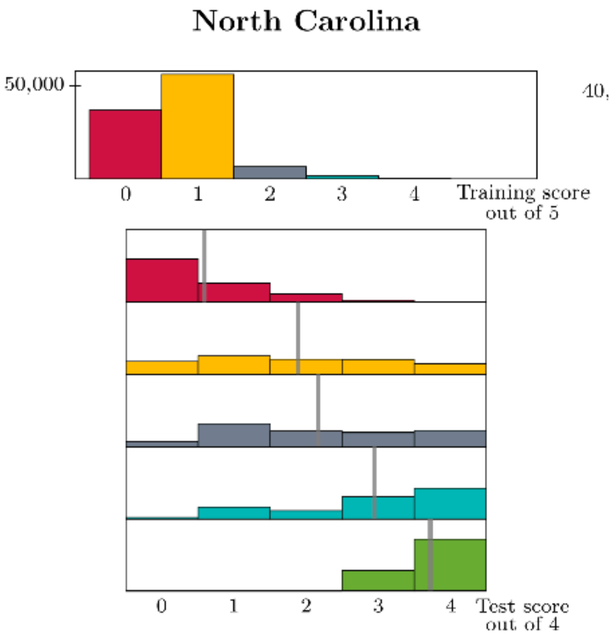
So only MA has a strong case to rebut the presumption of gerrymandering.

Can **people** pass the test?

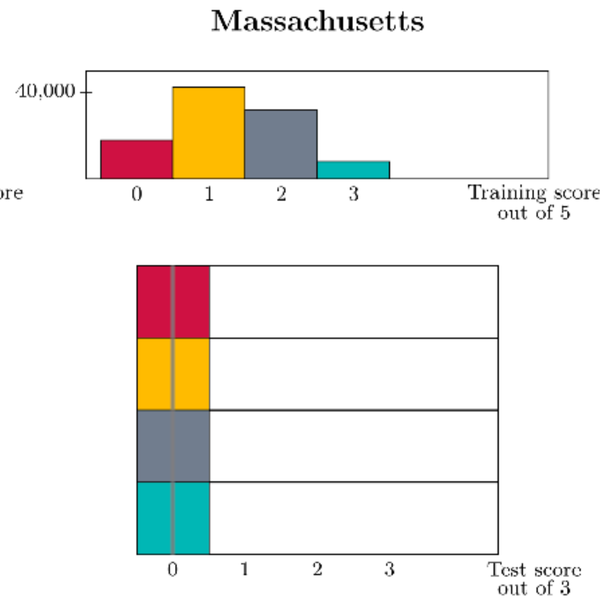
he did!



Does past proportionality predict future proportionality?



Training data: Gov08, Sen08, Sen10, Gov12, Pres12
Test data: Sen14, Pres16, Sen16, Gov16



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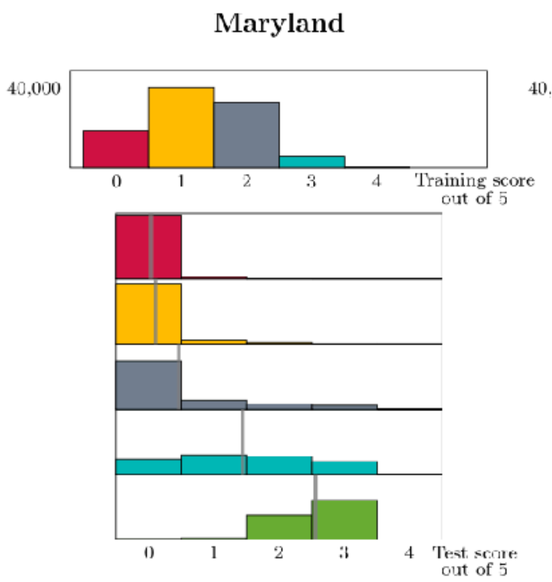
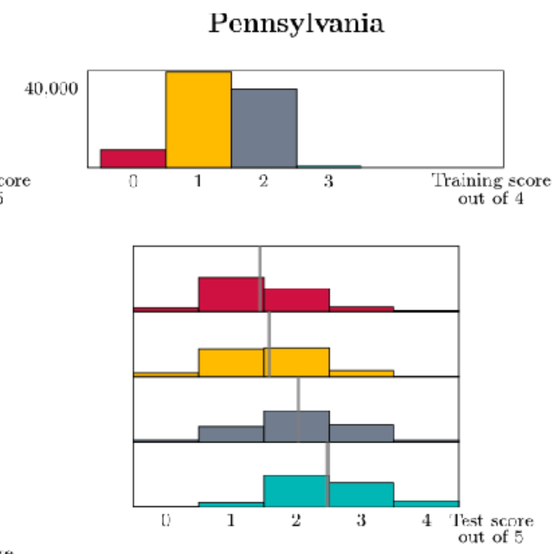
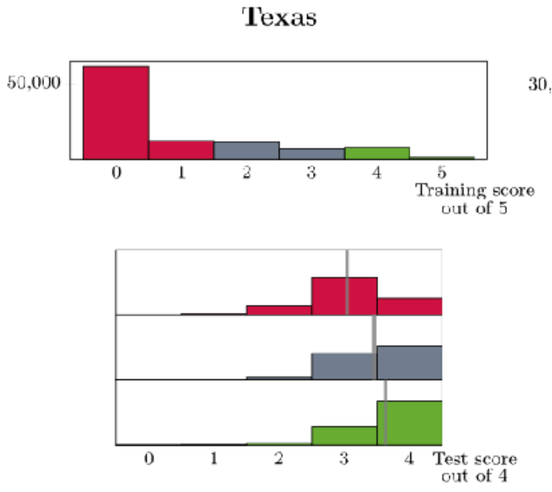
Test data: Pres16, Gov18, Sen18



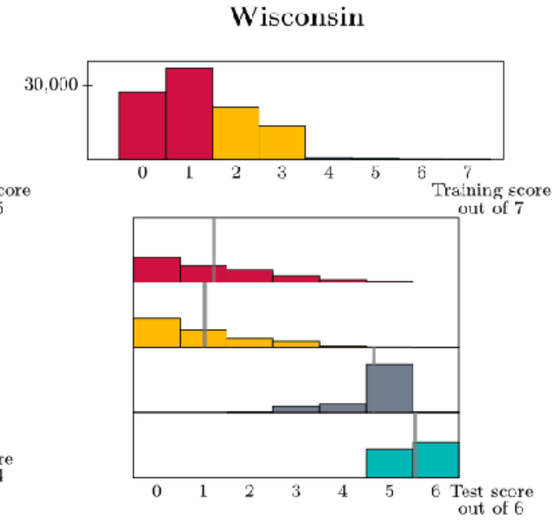
Training data: Pres12, Sen12, Gov14, AG14, Comp14
Test data: Pres16, Sen16, Sen18, AG18, Comp18



Training data: Gov10, Sen10, Sen12, Pres12
Test data: Sen16, Pres16, AG14, Gov14, AG16



Training data: Pres12, Sen12, Sen14, Gov14, AG14
Test data: Pres16, Sen18, Gov18, AG18



Training data: Pres12, Sen12, Gov12, Gov14, AG14, SOS14, Tre14
Test data: Pres16, Sen16, Sen18, Gov18, SOS18, Tre18

Putting it all together as a standard

Freedom to Vote proportionality standard

clear on what is measured ✅
sensitive to what matters ✅
passing test achievable ✅
creates good incentives ✅
allows healthy procedure 🆗
gets ground truth right ✅